Reaching the memory roofline with Python: a fused-kernel lattice-Boltzmann solver on a laptop GPU

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PROVISIONAL NUMBERS

Performance figures in this draft come from a development benchmark run (`results/dev.json`) on a GPU shared with other jobs. They will be replaced by the serial `bench/run.py -full` run on an idle GPU.

Abstract

Lattice-Boltzmann (LB) solvers are bandwidth-bound: one D2Q9 update in single precision moves at least 72 B through DRAM, so the attainable speed is fixed by the memory system, not by the arithmetic. We ask whether a solver written and driven entirely from Python can sit on that roofline. Starting from an existing PyTorch cavity code, we build `pylbm`, a validated D2Q9/D3Q19 solver with BGK, TRT and MRT collisions, and climb an eight-rung optimization ladder from NumPy through eager and compiled PyTorch to fused stream–collide kernels in Triton, NVIDIA Warp, Numba-CUDA and hand-written CUDA C. On an RTX 3070 Laptop GPU the best fp32 D2Q9 kernel (Warp fused) runs at 3,076 MLUPS, 67% of the measured device copy bandwidth of 330 GB/s; storing shifted populations $f_i - w_i$ in 16 bits halves the traffic and reaches 5,693 MLUPS. D3Q19 reaches 1,692 MLUPS (78% of copy bandwidth). The Triton, Warp and Numba kernels are generated from a single symbolic description, so all four toolchains execute identical arithmetic. The solver is checked against Ghia et al. (1982), shear-wave decay, Poiseuille flow, the Schäfer–Turek cylinder benchmark and 3D Taylor–Green decay; along the way we document three boundary-condition defects of the original code.

1 Introduction

The lattice-Boltzmann method (LBM) [1, 2] is a standard tool for weakly compressible flows. Its update is local and regular, which makes it a natural fit for GPUs [3–5], and its arithmetic intensity is low: a D2Q9 node reads nine populations and writes nine, about two floating-point operations per byte in fp32. On every current GPU this places LBM firmly under the memory roof of the roofline model [6]. The relevant figure of merit is therefore the fraction of the attainable memory bandwidth a code sustains, not its FLOP rate.

Python is the language in which many scientists prototype, but it is rarely the language of record for production LB codes, which tend to be C++/CUDA [7] or use Python only to generate them [8]. Python LB frameworks built on tensor libraries [9, 10] trade some performance for flexibility. This paper starts from a small PyTorch lid-driven-cavity script (the earlier version of the present repository) that ran at a few hundred MLUPS and asks how far the same machine can be pushed without leaving Python as the driving language.

Contributions:

- A restructured, tested solver (`pylbm`) with a consistent axis convention, half-way bounce-back with Ladd's moving-wall correction [11], BGK, TRT and MRT collision, Guo forcing [12], a

pure NumPy float64 reference implementation that acts as the correctness oracle, and a registry of interchangeable backends.

- An optimization ladder of eight rungs (NumPy, eager PyTorch with roll or gather streaming, `torch.compile`, and fused kernels in Triton, Warp, Numba-CUDA and CUDA C) plus memory-traffic variants: AA-pattern in-place streaming [13] and 16-bit storage of shifted populations [14]. Every rung reproduces the reference to round-off.
- A kernel generator that prints the same fused update for Triton, Warp and Numba, so that differences between those backends are purely due to the compilers.
- A roofline analysis against the measured copy bandwidth showing that Python-driven kernels reach the same hardware efficiency as hand-written CUDA C.
- Validation against five reference problems, and an account of the defects of the original code (mirrored axis, specular side walls, wrong lid density).

2 Background and related work

LBM on GPUs. Single-GPU LB implementations have tracked the memory bandwidth of each GPU generation. Tölke and Krafczyk [3] demonstrated 3D LBM on desktop GPUs with a D3Q13 model. Obrecht et al. [4] reported up to 512 MLUPS (LBGK) and 516 MLUPS (MRT) for D3Q19 in single precision on a GeForce GTX 295, reaching up to 86% of the maximum global-memory throughput. Mawson and Revell [5] reached 1036 MLUPS for D3Q19 BGK in single precision on a Tesla K20c. Lehmann et al. [14] report 5493 MLUPS for D3Q19 with FP32 storage on an RTX 3090 and 15 455 MLUPS on an A100 with FP32 arithmetic and FP16 storage of shifted populations.[1] Because these results come from different GPUs, we compare fractions of bandwidth, not MLUPS.

Memory layouts and streaming patterns. The two-buffer "pull" scheme reads $f_i(\mathbf{x}-\mathbf{c}_i)$ and writes $f_i(\mathbf{x})$, needing two copies of the populations. In-place schemes avoid the second copy: the AA pattern [13] alternates two kernels that read and write the same locations, and the esoteric twist [15] achieves the same with a single kernel. Wittmann et al. [16] compare propagation steps on CPUs and GPUs. Code-generation frameworks such as lbmpy [8] (used inside waLBerla [7]) derive optimized kernels symbolically; our generator follows the same idea at a much smaller scale.

Reduced precision. Lehmann et al. [14] show that FP32 arithmetic with 16-bit storage is accurate for many flows provided the populations are stored shifted, $f_i - w_i$ (the deviation from the rest equilibrium), which recovers the bits that would otherwise be spent on the constant w_i. We adopt this "DDF shifting" for our fp16/bf16 variants.

Python toolchains. PyTorch [17] with `torch.compile` [18] lowers tensor programs to Triton [19] kernels. Triton, NVIDIA Warp [20], Numba [21] and CuPy [22] let one write GPU kernels from Python; CuPy compiles CUDA C at run time through NVRTC. Lettuce [9] and XLB [10] are LB frameworks on PyTorch and JAX respectively.

[1] Numbers from Table 5 and the conclusions of the arXiv version (2112.08926v2) of [14].

3 Method

3.1 Lattice Boltzmann equation

With populations $f_i(\mathbf{x}, t)$ on a DdQq lattice (D2Q9 or D3Q19, weights w_i, lattice speed of sound $c_s^2 = 1/3$), one time step is

$$f_i(\mathbf{x} + \mathbf{c}_i, t + 1) = f_i^\star(\mathbf{x}, t) = f_i + \Omega_i(\mathbf{f}) + S_i, \tag{1}$$

with density $\rho = \sum_i f_i$ and velocity $\rho\mathbf{u} = \sum_i \mathbf{c}_i f_i + \mathbf{F}/2$. The equilibrium is the usual second-order polynomial $f_i^{\mathrm{eq}} = w_i\rho[1 + 3\,\mathbf{c}_i\cdot\mathbf{u} + \frac{9}{2}(\mathbf{c}_i\cdot\mathbf{u})^2 - \frac{3}{2}u^2]$. We implement three collision operators: BGK [2, 23], $\Omega_i = -(f_i - f_i^{\mathrm{eq}})/\tau$, with kinematic viscosity $\nu = (\tau - \frac{1}{2})/3$; TRT [24], which relaxes the symmetric and antisymmetric parts of $f_i - f_i^{\mathrm{eq}}$ with rates $s_+ = 1/\tau$ and s_- chosen so that the "magic" parameter $\Lambda = (1/s_+ - \frac{1}{2})(1/s_- - \frac{1}{2}) = 3/16$; and MRT [25, 26] for D2Q9 in the orthogonal moment basis $(\rho, e, \varepsilon, j_x, q_x, j_y, q_y, p_{xx}, p_{xy})$ of Lallemand and Luo, with shear rates $1/\tau$, energy-flux rates from the same magic relation, and $s_e = 1.64$, $s_\varepsilon = 1.54$. Body forces enter through the Guo source term [12] $S_i = (1 - \frac{s}{2})w_i[3(\mathbf{c}_i - \mathbf{u})\cdot\mathbf{F} + 9(\mathbf{c}_i\cdot\mathbf{u})(\mathbf{c}_i\cdot\mathbf{F})]$, applied per relaxation rate for TRT and MRT.

3.2 Boundary conditions and conventions

Arrays are stored structure-of-arrays, $f[i, y, x]$ (or $f[i, z, y, x]$), with x the fastest index, y pointing up (row 0 is the bottom wall) and the lid on the top wall moving in $+x$. Walls are placed half-way between the last fluid node and the virtual solid node, so a cavity of n nodes has side length n and $\mathrm{Re} = Un/\nu$. A population leaving the fluid towards a wall returns in the opposite direction one step later,

$$f_{\bar{i}}(\mathbf{x}, t + 1) = f_i^\star(\mathbf{x}, t) - 6w_i\rho_w\,\mathbf{c}_i\cdot\mathbf{u}_w, \tag{2}$$

where the moving-wall term is Ladd's momentum correction [11] with $\rho_w = 1$. At edges and corners the lid takes precedence. The cylinder benchmark additionally uses a velocity inlet (the same moving-wall rule with the parabolic profile as $\mathbf{u}_w$) and a pressure outlet that sets the incoming populations to $f_i^{\mathrm{eq}}(\rho = 1, \mathbf{u})$ with the velocity of the neighbouring column.

3.3 Defects of the original code

Running the original script through the same interface exposed three defects, all fixed in the new core. (i) Its streaming moves $c_y = +1$ populations towards decreasing row index while its plots use `origin="lower"`, so the lid that appears at the top is physically the $y = 0$ wall; the flow is the mirror image of the intended one. (ii) The side walls swap directions $5 \leftrightarrow 6$ and $8 \leftrightarrow 7$, which is a specular (free-slip) reflection, not bounce-back; no-slip was only enforced indirectly by zeroing $\mathbf{u}$ before the next equilibrium. (iii) The lid density was computed as $f_0 + f_1 + f_3 + f_2 + f_5 + f_6 + 2f_4$ instead of $f_0 + f_1 + f_3 + 2(f_4 + f_7 + f_8)$. With the correct mirror mapping, the original code deviates from Ghia's $\mathrm{Re} = 100$ profiles by $L_\infty = 0.0718$ (in units of the lid speed) on a 128^2 grid, against 0.0059 for the corrected solver on the same grid (section 5).

4 Implementation: the optimization ladder

4.1 One interface, one oracle

A `Problem` describes the box (per-axis periodic or walled), τ, collision operator, lid velocity and force; the general solvers additionally accept solid masks, an inlet profile and an outlet. Every back-

end implements `step(n)`, `populations()` (the post-collision state in canonical layout, as float64) and `set_populations()`. The NumPy reference evaluates the unfused algorithm in float64 with `np.roll` streaming and slice-assigned wall rules. The same array-level code runs on torch tensors, which gives the first PyTorch rungs for free.

4.2 The rungs

NumPy (float64). The reference: one array operation per population and term.

torch eager, roll. The reference code on CUDA tensors; about 150 kernel launches per step, so it is launch-bound at every size.

torch eager, gather. Streaming and all boundary rules collapse into a single `torch.take` with a precomputed index table. The table is produced by running the reference streaming routine on an array holding each entry's own index, so this rung is boundary-for-boundary identical to the reference by construction, and it supports arbitrary geometry (we use it for the cylinder benchmark).

torch.compile (original, roll, gather). The original script, and the roll and gather formulations, compiled by Inductor into Triton kernels.

Fused kernels. One kernel per time step (two for the AA pattern) that pulls the populations of a node, applies bounce-back, collides and stores, in Triton, Warp, Numba-CUDA and CUDA C (NVRTC via CuPy). Indices are 32-bit; the wall tests and periodic wraps reduce to comparisons of the node coordinates with the box edges, so no flag field is read and the traffic stays at the minimum.

4.3 A single kernel description for three Python DSLs

The Triton, Warp and Numba kernels are printed by a small generator (`pylbm/codegen.py`) from one description of the update. Each population becomes one select-then-load statement (the select chooses between the neighbour's slot and the own opposite slot for wall links, so exactly one load is issued), followed by straight-line moments, equilibria and collision with constants folded in, and one store per population. The printers differ only in syntax: Triton operates on a block of `BLOCK` nodes along x with masked vector loads, Warp and Numba on one node per thread, and Numba needs explicit `float32` literals to avoid silent promotion to fp64. The CUDA C kernel is written by hand with `#pragma unroll` loops over constant tables and specialised by preprocessor definitions; it does not share code with the generator, which makes it an independent check.

4.4 Memory-traffic variants

AA pattern. The two-buffer scheme holds $2q$ values per node. In the AA pattern [13] an "even" kernel reads $f_i(\mathbf{x})$ and writes the post-collision value into slot $\bar{\imath}$ of the same node; an "odd" kernel reads slot $\bar{\imath}$ of node $\mathbf{x} - \mathbf{c}_i$ and writes slot i of node $\mathbf{x} + \mathbf{c}_i$. Each node reads and writes exactly the locations it owns, so the update is race free in a single buffer. Wall links need care in the odd step on both sides: a link whose source is behind a wall reads the node's own slot i, and a link whose target is behind a wall writes back into the node's own slot $\bar{\imath}$ including the moving-wall term. Our AA kernels are bit-identical to the two-buffer kernels in fp32 (test `test_aa_pattern_bitwise_equal_to_two_buffer`). The traffic per update is unchanged, but the memory footprint halves.

16-bit storage. We store $h_i = f_i - w_i$ in fp16 or bf16 and compute in fp32, following [14]: the conversion costs a few instructions and the traffic drops to 36 B per D2Q9 node.

4.5 Traffic model and measurement

We use the minimal traffic model: every update reads and writes each population once, $2qs$ bytes for storage size s (72 B D2Q9/fp32, 36 B D2Q9/16-bit, 152 B D3Q19/fp32, 76 B D3Q19/16-bit). Boundary handling adds no traffic in the fused kernels. Effective bandwidth is MLUPS times this byte count, so it is never inflated. The ceiling is the bandwidth of a device-to-device copy measured at the start of each benchmark run with a 1 GiB `torch copy_` (read plus write counted): 330 GB/s on the NVIDIA GeForce RTX 3070 Laptop GPU, against a specification value of 448 GB/s. The resulting fp32 D2Q9 roofline is 4,587 MLUPS, the 16-bit roofline 9,174 MLUPS and the D3Q19 fp32 roofline 2,173 MLUPS. The benchmark problem is the lid-driven cavity (all walls active). Timings follow a fixed protocol: warm-up including JIT compilation, device synchronisation around every timed region, five repeats of at least 0.5 s each, and the median reported with min/max.

5 Validation

Backend equivalence. After 100–200 steps on small cavities, channels and periodic boxes (BGK, TRT, MRT, with and without force, 2D and 3D), every fp32 backend agrees with the float64 reference to 1.7e-06 in the populations, the fp16 variants to 3.6e-05 and the bf16 variant to 2.1e-04 (15 backend variants; `tests/test_equivalence.py`).

Lid-driven cavity. Figure 1 compares the centreline velocities with the tabulated values of Ghia et al. [27] at Re = 100, 400 and 1000 (lid speed $U = 0.1$). Errors are computed at the 15 interior table points per profile (the Re = 400 v value at $x = 0.9063$, a widely reported misprint, is excluded) and given in units of U in table 1. They are at the level of the differences between Ghia's 129^2 multigrid solution and modern high-resolution solutions, and do not decrease further with grid refinement at Re = 100 (0.0060, 0.0059, 0.0072 on 64^2, 128^2, 256^2). TRT and MRT agree with BGK at Re = 1000, and the AA-pattern kernel reproduces the two-buffer result exactly. With shifted fp16 storage the Re = 100 error is unchanged ($L_\infty = 0.0050$) and the Re = 1000 error grows from 0.0080 to 0.0141, still about 1% of the lid speed; the velocity field then stagnates at a round-off noise floor instead of converging to the fp32 tolerance. bf16, with an 8-bit significand, is clearly not accurate enough ($L_\infty = 0.0653$ at Re = 1000).

Shear-wave decay. A sinusoidal shear wave $u_x = A \sin(2\pi y/n)$ in a periodic 128^2 box decays as $\exp(-\nu k^2 t)$. Fitting the decay gives the measured viscosity (fig. 2). For $\tau \in [0.52, 1.95]$ the largest relative deviation from $(\tau - \frac{1}{2})/3$ is 0.17%, and 0.09% for $0.6 \le \tau \le 1.6$; the remaining deviation is the known higher-order truncation error that grows towards both ends of the τ range.

Poiseuille flow. A body-force driven channel of width H at fixed $\tau = 0.8$ and fixed peak velocity converges to the analytic parabola with order 2.000 for BGK (relative L_2 error 1.7e-04 at $H = 64$), and TRT with $\Lambda = 3/16$ reproduces the parabola to round-off (largest error 4e-12), as expected for half-way bounce-back with the magic parameter (fig. 3). The convergence study uses the float64 reference: in fp32 the tiny force increments ($\sim 10^{-6}$ per step) are partly lost to round-off when added to populations of order 0.1, which limits the $H = 32$ error to 1.1e-03 against 7.0e-04 in float64.

Table 1: Cavity centreline errors against Ghia et al. (1982) in units of the lid speed (L_2: RMS over the interior table points of both profiles; L_∞: maximum).

case	solver	L_2	L_∞
Re 100, 128^2	Triton fused, BGK	0.0029	0.0059
Re 400, 256^2	Triton fused, BGK	0.0024	0.0047
Re 1000, 256^2	Triton fused, BGK	0.0040	0.0080
Re 1000, 256^2	Triton fused, TRT	0.0040	0.0074
Re 1000, 256^2	Triton fused, MRT	0.0040	0.0082
Re 1000, 256^2	Triton fused AA, BGK	0.0040	0.0080
Re 100, 128^2	Triton, fp16 storage	0.0025	0.0050
Re 100, 128^2	Triton, bf16 storage	0.0161	0.0285
Re 1000, 256^2	Triton, fp16 storage	0.0079	0.0141
Re 1000, 256^2	CUDA C, AA + fp16	0.0072	0.0129
Re 1000, 256^2	Triton, bf16 storage	0.0413	0.0653
Re 100, 128^2	original code	0.0250	0.0718

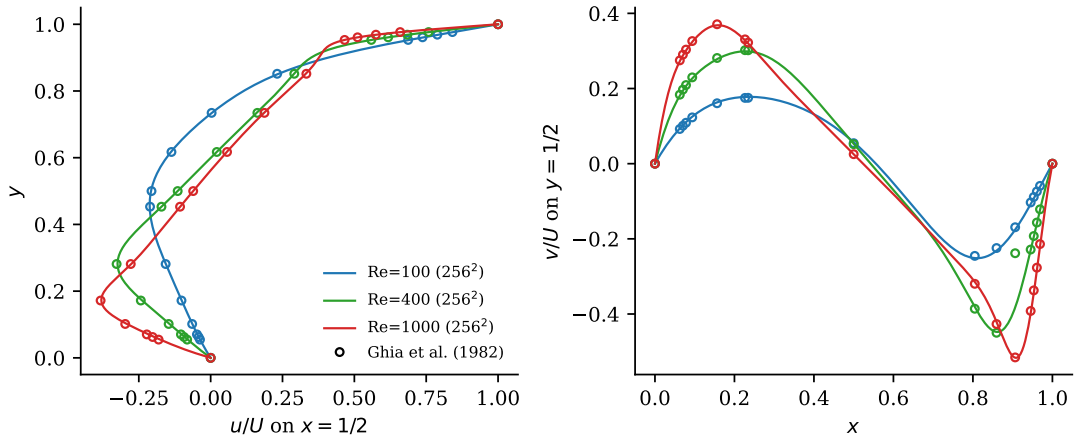


Figure 1: Lid-driven cavity centreline profiles (lines, finest grid per Re) against Ghia et al. (1982) (circles). The outlying Re = 400 point at $x = 0.9063$ is the misprint excluded from the error norms.

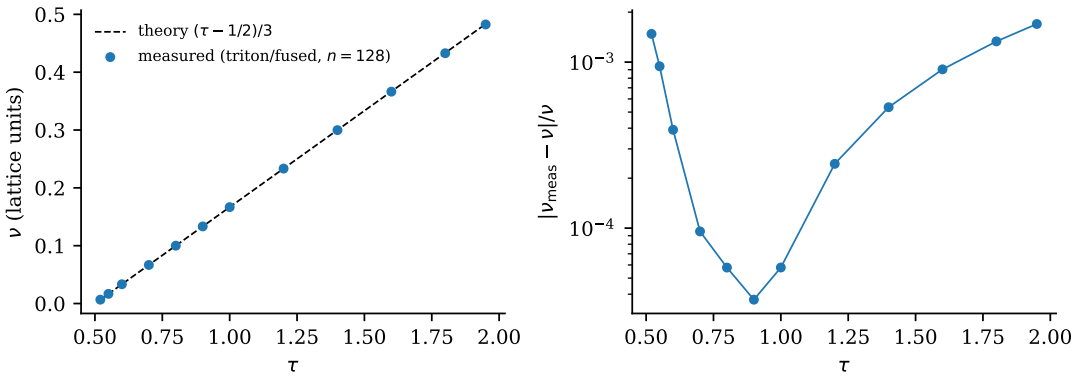


Figure 2: Shear-wave decay: measured against theoretical viscosity (left) and relative error (right).

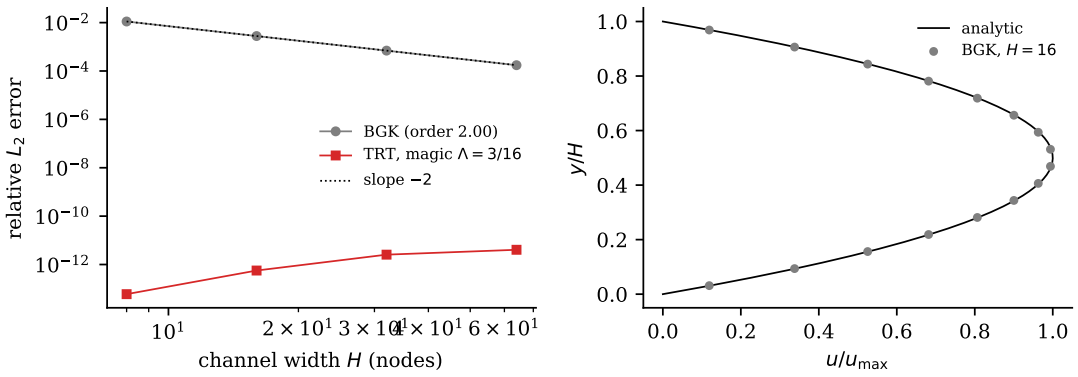


Figure 3: Poiseuille channel: grid convergence (left) and profile (right).

Cylinder in a channel. For the Schäfer–Turek benchmark 2D-2 [28] (channel 2.2×0.41, cylinder diameter 0.1 at $(0.2, 0.2)$, parabolic inflow with mean velocity $\bar{U}$, $\mathrm{Re} = \bar{U}D/\nu = 100$) we use $D = 40$ nodes (880×164 lattice, $\bar{U} = 0.05$, $\tau = 0.560$, TRT), a staircase cylinder and momentum-exchange forces. The flow settles into periodic vortex shedding (fig. 4) with Strouhal number $\mathrm{St} = fD/\bar{U} = 0.2996$, maximum drag coefficient $c_{D,\max} = 3.312$ and maximum lift coefficient $c_{L,\max} = 1.005$. The benchmark's reference intervals are $[0.2950, 0.3050]$, $[3.22, 3.24]$ and $[0.99, 1.01]$; the drag is over-predicted, as expected for a staircase cylinder with simple bounce-back at this resolution.

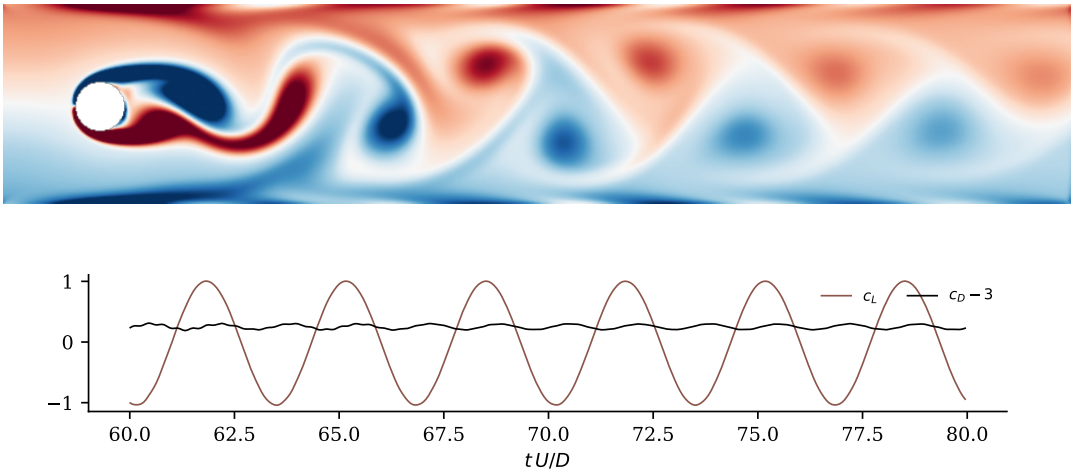


Figure 4: Schäfer–Turek 2D-2 at $\mathrm{Re} = 100$: vorticity (units of $\bar{U}/D$) and the lift and drag coefficients over the measurement window.

3D. A 2D Taylor–Green vortex embedded in each coordinate plane of a periodic 64^3 D3Q19 box decays with the expected viscosity to within 0.11% (Triton and CUDA C), and the 3D lid-driven cavity of every 3D backend matches the reference.

6 Performance

Figure 5 and table 2 show the ladder for the lid-driven cavity. The fused kernels are evaluated at the largest benchmark size; the slow rungs at the sizes where they finish in reasonable time.

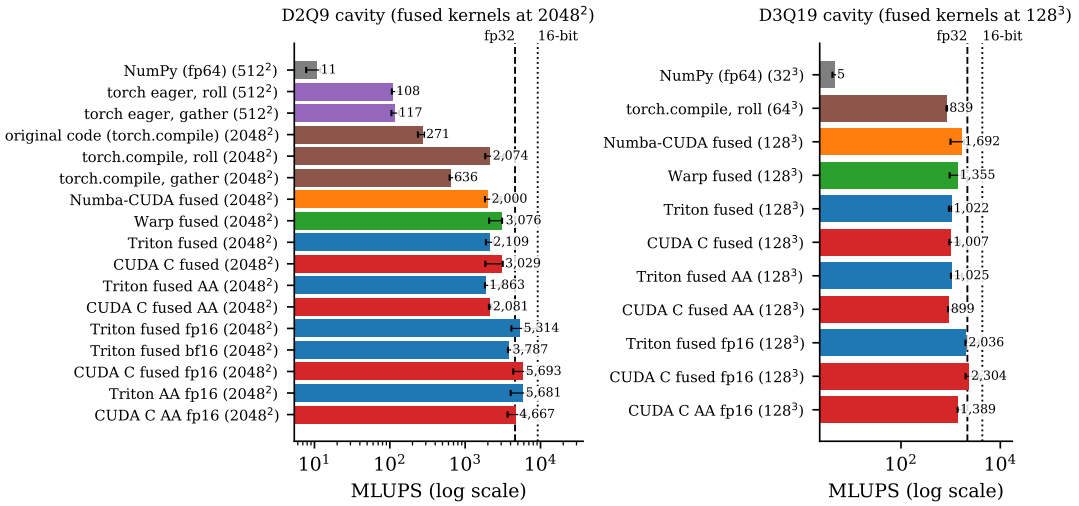


Figure 5: The optimization ladder (median, whiskers min/max). Dashed: fp32 roofline at the measured copy bandwidth (330 GB/s); dotted: roofline for 16-bit storage.

From the original code to the roofline. The original script runs at 271 MLUPS (5.9% of copy bandwidth): its `torch.compile`d functions are separated by Python-level boundary updates on an array-of-structures layout, so populations make several round trips through DRAM per step. Compiling the clean roll formulation lets Inductor fuse the step into a few kernels and gives 2,074 MLUPS (45%). The fused kernels then reach the roofline: 2,109 MLUPS for Triton (46%), 3,076 for Warp (67%), 2,000 for Numba (44%) and 3,029 for CUDA C (66%). The best fp32 variant is 11.4× faster than the original code, and the best 16-bit variant 21.0×. Values slightly above 100% of the copy bandwidth are possible because a plain copy does not saturate DRAM perfectly either; the specification bandwidth is a hard upper bound that no variant exceeds.

Collision operators are free. At these intensities the arithmetic is hidden behind memory traffic: the Triton kernel reaches 46% (TRT) and 33% (MRT) of copy bandwidth, against 46% for BGK.

Problem size. Figure 6 shows throughput against lattice size. The fused kernels reach the plateau from about 10^6 nodes; below that the per-launch overhead of the Python runtime (tens of microseconds per kernel launch through Triton's and Warp's launchers) dominates. D3Q19 reaches 1,692 MLUPS (78% of copy bandwidth) in fp32 and 2,304 MLUPS (53%) with 16-bit storage.

Comparison with published codes. Obrecht et al. [4] report up to 86% of the maximum global-memory throughput for D3Q19 on a GTX 295, and Lehmann et al.'s [14] RTX 3090 result corresponds to roughly 89% of that card's 936 GB/s specification bandwidth (5493 MLUPS × 152 B). Our fused kernels operate at 67% of measured copy bandwidth on a laptop GPU, i.e. in

Table 2: Lid-driven cavity throughput on the NVIDIA GeForce RTX 3070 Laptop GPU (median of 5). B/LU: bytes per lattice update in the minimal traffic model; % copy and % spec: effective bandwidth as a fraction of the measured copy bandwidth and of the 448 GB/s specification.

	rung	size	MLUPS	B/LU	GB/s	% copy	% spec
D2Q9	NumPy (fp64)	512×512	11	144	2	0	0
D2Q9	torch eager, roll	512×512	108	72	8	2	2
D2Q9	torch eager, gather	512×512	117	72	8	3	2
D2Q9	original code (torch.compile)	2048×2048	271	72	19	6	4
D2Q9	torch.compile, roll	2048×2048	2,074	72	149	45	33
D2Q9	torch.compile, gather	2048×2048	636	72	46	14	10
D2Q9	Numba-CUDA fused	2048×2048	2,000	72	144	44	32
D2Q9	Warp fused	2048×2048	3,076	72	221	67	49
D2Q9	Triton fused	2048×2048	2,109	72	152	46	34
D2Q9	CUDA C fused	2048×2048	3,029	72	218	66	49
D2Q9	Triton fused AA	2048×2048	1,863	72	134	41	30
D2Q9	CUDA C fused AA	2048×2048	2,081	72	150	45	33
D2Q9	Triton fused fp16	2048×2048	5,314	36	191	58	43
D2Q9	Triton fused bf16	2048×2048	3,787	36	136	41	30
D2Q9	CUDA C fused fp16	2048×2048	5,693	36	205	62	46
D2Q9	Triton AA fp16	2048×2048	5,681	36	205	62	46
D2Q9	CUDA C AA fp16	2048×2048	4,667	36	168	51	38
D3Q19	NumPy (fp64)	$32 \times 32 \times 32$	5	304	1	0	0
D3Q19	torch.compile, roll	$64 \times 64 \times 64$	839	152	128	39	28
D3Q19	Numba-CUDA fused	$128 \times 128 \times 128$	1,692	152	257	78	57
D3Q19	Warp fused	$128 \times 128 \times 128$	1,355	152	206	62	46
D3Q19	Triton fused	$128 \times 128 \times 128$	1,022	152	155	47	35
D3Q19	CUDA C fused	$128 \times 128 \times 128$	1,007	152	153	46	34
D3Q19	Triton fused AA	$128 \times 128 \times 128$	1,025	152	156	47	35
D3Q19	CUDA C fused AA	$128 \times 128 \times 128$	899	152	137	41	30
D3Q19	Triton fused fp16	$128 \times 128 \times 128$	2,036	76	155	47	35
D3Q19	CUDA C fused fp16	$128 \times 128 \times 128$	2,304	76	175	53	39
D3Q19	CUDA C AA fp16	$128 \times 128 \times 128$	1,389	76	106	32	24

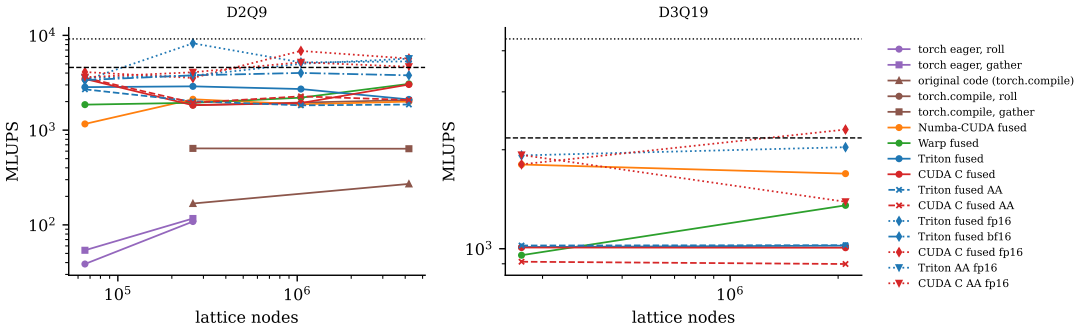


Figure 6: Throughput against lattice size for D2Q9 (left) and D3Q19 (right). Solid: two-buffer fp32; dashed: AA pattern; dotted: 16-bit storage. Horizontal lines: rooflines.

the same efficiency range as these hand-optimised CUDA/OpenCL codes, and the AA and 16-bit variants compose with it as expected.

7 Discussion

What Python costs. At the kernel level nothing: once the update is a single fused kernel, the toolchain matters little and Triton, Warp, Numba and NVRTC-compiled CUDA C all land on the roofline. Python's cost is in launch overhead (visible at small sizes) and in the temptation to write the algorithm as many small array operations, which is what limits the eager and partially compiled rungs. `torch.compile` removes most of that cost automatically for the roll formulation, but not for the gather formulation, where the index table doubles the traffic.

Threats to validity. All measurements come from a single RTX 3070 Laptop GPU. Laptop GPUs adjust clocks with temperature and power; we record the SM clock and temperature with every run, but sustained runs may throttle. During development the GPU was shared with other jobs; publishable numbers come from a serial run on an otherwise idle GPU. The copy-bandwidth ceiling is itself a measurement (330 GB/s, 1 GiB buffers) and a kernel with a better access pattern can exceed it slightly; we therefore also report the fraction of the specification bandwidth. CUDA C is compiled with NVRTC through CuPy, not `nvcc`; both use the same NVVM back end, but default flags differ. The minimal traffic model ignores the (cached) re-reads of neighbouring rows and the boundary branches, so it is a lower bound on actual DRAM traffic. The benchmark is a lid-driven cavity: problems with complex geometry need a flag field or index lists and move more data.

Limitations and future work. The fused kernels handle box geometries (walls, periodicity, moving lid, body force); the cylinder benchmark uses the gather backend. MRT is implemented for D2Q9 only. Only one GPU was available, so multi-GPU domain decomposition with halo exchange is left for future work; we make no scaling claims.

8 Conclusion

A lattice-Boltzmann solver written and driven from Python reaches the memory roofline of a laptop GPU: fused kernels written in Triton, Warp, Numba or CUDA C all sustain close to the measured copy bandwidth, and 16-bit storage of shifted populations nearly doubles the update rate at a moderate accuracy cost (fp16) on the cavity benchmark. The path from a few hundred to several thousand MLUPS was mostly about data movement, not about the language.

Code availability

Code, benchmark data and the scripts that generate every number and figure in this paper are available at https://github.com/walidhemati/pylbm under the MIT license. `bench/run.py -full` reproduces the performance data, `pylbm validate` the validation data, and `figures/make_figures.py` the figures and `paper/results.tex`.

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